

Ports — How Data Finds the Right Service

Study notes

1. What Is a Port?

Every device on a network has one IP address, but that same device can run many services at once — a website, an email server, a remote login session, a database. A **port** is a number that tells incoming data exactly which service on that device it's meant for. If the IP address is a building's street address, the port is the specific apartment number inside it.

Port numbers range from 0 to 65535 — 65,536 possible "doors" on a single device. They fall into three ranges:

- **0–1023 — Well-known ports.** Reserved for standard, widely used services (HTTP, HTTPS, SSH, DNS, and so on).
- **1024–49151 — Registered ports.** Used by specific applications (for example, MySQL on 3306).
- **49152–65535 — Dynamic/private ports.** Temporarily assigned by your own device when it makes an outgoing connection.

2. TCP vs UDP, Briefly

A port number is always paired with a transport protocol — almost always TCP or UDP.

- **TCP (Transmission Control Protocol)** — checks that data arrives complete and in order. Used when accuracy matters more than speed: web pages, file transfers, logins.
- **UDP (User Datagram Protocol)** — sends data fast, without checking delivery. Used when speed matters more than perfection: video calls, live streaming, gaming.

Some services, like DNS, use both — depending on the situation (see the note under the table below).

3. The 5 Ports You Must Know

Port	Service	Full Form	Protocol	What It's Used For
80	HTTP	HyperText Transfer Protocol	TCP	Loading normal, unencrypted websites.
443	HTTPS	HTTP Secure	TCP	Loading secure, encrypted websites — the padlock icon in your browser.
22	SSH	Secure Shell	TCP	Securely logging into a remote server or device via command line.
53	DNS	Domain Name System	TCP & UDP	Translating a domain name (google.com) into its IP address.
3389	RDP	Remote Desktop Protocol	TCP	Remotely accessing a Windows computer's full desktop.

Note: DNS mainly uses UDP for fast lookups, and switches to TCP for larger responses, such as zone transfers between DNS servers.

4. More Common Ports Worth Knowing

Port	Service	Protocol	What It's Used For
21	FTP	TCP	Transferring files to and from a server.
23	Telnet	TCP	Old, unencrypted remote login — rarely used today, but still shows up in interview questions about legacy protocols.
25	SMTP	TCP	Sending email between mail servers.

Port	Service	Protocol	What It's Used For
110	POP3	TCP	Downloading email to a single device.
143	IMAP	TCP	Syncing email across multiple devices.
445	SMB	TCP	Windows file and printer sharing.
3306	MySQL	TCP	Connecting to a MySQL database.
8080	HTTP (alt)	TCP	Common alternate port for web servers and proxies.

5. What Actually Happens: A Real Example

Say you type <https://example.com> into your browser. Here's the real sequence:

- 1 Your device asks a DNS server (port 53) to translate example.com into an IP address.
- 2 Your device opens a TCP connection to that IP address on port 443 (HTTPS).
- 3 A secure handshake happens — your browser and the server agree on encryption.
- 4 The webpage data travels back over that same port 443 connection.

Almost everything you do online comes down to a handful of these well-known ports doing the actual work.

6. Quick Interview Q&A

Q: What port range counts as "well-known" ports?

A: 0–1023. These are reserved for standard, widely used services.

Q: Which port does HTTPS use, and why is it different from HTTP's port?

A: 443. HTTPS adds encryption (TLS/SSL) on top of HTTP, so it needs its own dedicated port, separate from plain HTTP (80).

Q: Why does DNS use both TCP and UDP?

A: UDP (port 53) handles regular, fast lookups. TCP (port 53) is used for larger responses, such as zone transfers between DNS servers.

Q: What's the practical difference between SSH and RDP?

A: SSH (22) gives a command-line connection, commonly used on Linux servers. RDP (3389) gives a full graphical desktop, used mainly on Windows machines.

7. Try It Yourself

Want to see this in action on your own device?

- **Windows:** open Command Prompt and type **netstat -an**
- **Mac/Linux:** open Terminal and type **netstat -an** or **lsof -i**

This lists every port currently in use on your computer, and which ones are actively listening for connections.

8. Memory Hook

1 computer = 65,536 doors (ports). Finding the right service = finding the right door. The 5 core ports above are the ones you'll use and get asked about the most.